

MATHEUS LAJARIN ZEM

Full Stack Developer

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PROFESSIONAL SUMMARY

Full Stack Developer who delivers end to end — from the data model and the API to the frontend, infrastructure and production deployment. Computer Engineering student, working both on large-scale enterprise products and on complete personal projects.

I value professional quality: code versioned in a DEV/HMG/PRD workflow, controlled database migrations, CI/CD pipelines, staging environments before production, and security (JWT authentication and encryption of sensitive data). I have real flexibility with technologies and choose the right tool for each need — I started solid in Java/Spring, adopted Python when the problem called for productivity and a data/automation ecosystem, and moved to Go (Golang) when the focus was performance and runtime simplicity. I learn and switch stacks to fit the problem, not the other way around.

I incorporate AI assistants (KIMI and Claude) simultaneously into my development workflow — speeding up coding, code review, debugging and architecture decisions, always validating the result.

PROFESSIONAL EXPERIENCE

TOTVS — Mid-Level Full Stack Developer (B2B Contractor) Nov 2024 – present

Development and maintenance of management platforms for the agro-industrial sector built on a microservices architecture (Spring Boot, Spring Cloud Gateway and Eureka) with Angular web portals and PostgreSQL.

- **Microservices:** Work on systems composed of dozens of microservices for managing costs, cash flow, materials, suppliers, weighing/scale and production processes.
- **APIs & integrations:** Development of REST APIs, service-to-service integrations and integration interfaces with external systems and client ERPs.
- **Database:** Modeling, table creation and PostgreSQL query optimization for performance and operational stability.
- **Quality & versioning:** Debugging and tuning of services, APIs and integrations, ensuring stability in the client environment; version control with Git in a DEV / HMG / PRD workflow.

CNH Industrial — Project Management Intern Jun 2022 – Jun 2024

Parts management across projects, from registration to supplier relations. Built responsive spreadsheets, macros and pivot tables in Excel to optimize control and data analysis.

Elring Klinger — Kaizen Intern Mar 2022 – May 2022

Tracking of Kaizen requests from concept to implementation. Developed macros to automate report generation for various departments.

PERSONAL PROJECTS

Full stack applications I built individually, from backend to frontend, infrastructure and deployment, using AI (KIMI and Claude in parallel) as a pair-programming partner for architecture, implementation and code review.

- **Finance** — Personal finance management platform (Monevas product). Java 21 + Spring Boot, Angular + PrimeNG, PostgreSQL, with CI/CD and deployment on my own VPS — detailed below.
- **Educa** — Education platform. Spring Boot backend and Angular 21 frontend (PrimeNG, TailwindCSS, Chart.js, Quill editor).
- **Monitorame** — Monitoring and location tracking. Spring Boot backend, Angular frontend with maps (Leaflet) and a Flutter mobile app.
- **DB Modeler** — Visual database modeler in Go (Gin): reverse and forward SQL engineering with generation of DDL, DBML and Mermaid (ER) diagrams.

- **Device Manager** — Device management and location tracking. Java 21 + Spring Boot backend and cross-platform Flutter frontend.
- **Git Manager** — Local Git repository manager. Java backend (Javalin + JGit) and Flutter Desktop frontend.
- **Aprenda** — My own technical course track (Linux, Docker, Go, Python, SQL/PostgreSQL, Frontend, AWS, Kubernetes, Networking, Security, IoT, Java and DevOps).

HANDS-ON PERSONAL PROJECT EXAMPLE

Finance (Monevas) — Personal finance management platform *hmg.monevas.com*

A complete personal finance product, conceived, developed and operated entirely by me — backend, frontend, database, CI/CD and my own infrastructure.

- **Backend:** Java 21 + Spring Boot 3.3.5 as a modular Maven monolith (auth, users, bank accounts/transactions, investments, planning, reports, notifications and AI modules), exposing REST APIs.
- **Frontend:** Angular + PrimeNG with internationalization (pt-BR, English and Spanish) and per-environment builds.
- **Data:** PostgreSQL with versioned migrations via Flyway (Hibernate in validate mode, no auto-DDL), ensuring predictable schema evolution.
- **Security:** Spring Security with JWT authentication (JJWT), roles/permissions and encryption of sensitive data.
- **Integrations:** Integration with a financial market API (brapi) for stock quotes and asset data.
- **CI/CD:** GitHub Actions building the Docker image and publishing to the GitHub Container Registry (GHCR); frontend auto-deployed on Vercel.
- **Infra & deploy:** Personal Debian VPS with Docker. Backend and PostgreSQL run as containers on a shared Docker network, with Watchtower automatically updating the published image, healthcheck via Spring Actuator and rotated logs. Real staging environment at hmg.monevas.com / api-hmg.monevas.com before promoting to production.
- **Challenges solved:** Setting up the infrastructure from scratch (Docker network, ports exposed only on localhost behind a proxy, per-environment CORS, secret management via .env kept out of version control), debugging inter-container communication (host resolution by service name, not localhost) and maintaining the dev/hmg/prd separation to validate every change before it goes live.

TECHNICAL SKILLS

Languages: Java (advanced), Python (intermediate), TypeScript/JavaScript, Dart, Go, SQL

Backend: Spring Boot, Spring Cloud (Gateway, Eureka), REST APIs, JWT, microservices, Javalin, Django REST Framework, Gin

Frontend: Angular, PrimeNG, TailwindCSS, RxJS, Flutter (mobile / desktop / web)

Database: PostgreSQL — modeling, table creation and query optimization

DevOps & Infra: Git (DEV/HMG/PRD), Docker and Docker Compose, CI/CD (GitHub Actions, GHCR), Vercel, Debian/Linux VPS, Flyway, Maven

AI in development: Simultaneous use of KIMI and Claude for coding, code review, debugging and architecture

EDUCATION

Computer Engineering — Escola de Engenharia de Piracicaba (EEP) *expected Dec 2026*

Evening program.

LANGUAGES

- **English** — Advanced; courses at Red Balloon, Cultura Inglesa and Wise Up; hands-on practice during trips to the USA.
- **Portuguese** — Native.
- **Spanish** — Basic / professional working knowledge.